ANALYSIS

Multi-Agent Invoice Reconciliation System

1. Where does OCR/extraction fail? How do your agents compensate?

Common Failure Points:

- Low-resolution scans: Gemini Vision struggles with blurry text, particularly small font sizes in line item tables where unit prices blend together

- Rotated/skewed documents: While the model handles slight rotations, 45°+ angles cause field misalignment and partial extraction

- Handwritten annotations: Stamps, handwritten notes, and corrections over printed text create ambiguity the model may extract the wrong value or both

- Non-standard table formats: Invoices without clear grid lines or with merged cells cause line item parsing errors, especially quantity/price column confusion

- Currency symbol variations: £, GBP, and numeric-only totals sometimes parse inconsistently

Agent Compensation Strategies:

The Document Intelligence Agent implements several fallback mechanisms:

1. Confidence scoring: Each extraction includes a `confidence_assessment` (high/medium/low). Low-confidence extractions trigger the `warnings` field, alerting downstream agents

2. Field validation: Missing critical fields (invoice_number, total) are flagged immediately rather than passing garbage data forward

3. Graceful degradation: If line items fail to parse, the agent still extracts header-level data (supplier, total, dates) enabling partial matching

4. Resolution Agent escalation: When extraction confidence drops below 70%, the system automatically recommends `escalate_to_human` rather than making unreliable decisions

The Matching Agent further compensates by not relying solely on extracted PO references it implements fuzzy supplier and product matching as fallbacks when OCR produces invalid PO numbers.

2. How would you improve accuracy from 70% to 95%?

Immediate Improvements (70% → 85%):

1. Multi-model ensemble: Run Gemini Vision AND Claude Vision in parallel, cross-validate extractions, flag disagreements for review

2. Pre-processing pipeline: Add OpenCV-based deskewing, contrast enhancement, and noise reduction before vision model processing

3. Template detection: Build a classifier to identify common invoice templates, then use template-specific extraction prompts with known field positions

4. Structured output enforcement: Use Gemini's JSON mode with strict schema validation to reduce parsing failures

Advanced Improvements (85% → 95%):

5. Fine-tuned extraction model: Train a specialized model on 10,000+ annotated pharmaceutical invoices to learn domain-specific patterns

6. Human-in-the-loop feedback: Implement the bonus "Human Reviewer Agent" that captures correction patterns and dynamically adjusts extraction prompts

7. Multi-pass extraction: First pass extracts structure, second pass validates numbers mathematically (qty × price = line_total, sum of lines = subtotal)

8. Confidence calibration: Train a separate model to predict extraction accuracy based on document quality metrics, enabling smarter routing

Key insight: The jump from 70% to 85% is engineering (better preprocessing, prompts). The jump from 85% to 95% requires ML investment (fine-tuning, calibration).

3. How would you validate this system at 10,000 invoices/day scale?

Validation Strategy:

1. Shadow mode deployment: Run the agent system in parallel with existing manual process for 2 weeks. Compare every decision without acting on agent recommendations. Measure:

- Extraction accuracy (field-level F1 scores)
- Matching precision/recall
- False positive rate for discrepancies
- Decision agreement rate with human reviewers

2. Stratified sampling: At 10K/day, review 1% (100 invoices) daily via random sampling stratified by:

   - Confidence score buckets (high/medium/low)

   - Decision type (auto-approve vs escalate)

   - Supplier diversity

   - Document quality scores

3. Automated regression testing: Maintain a golden dataset of 500 annotated invoices. Run nightly to catch model drift or code regressions. Alert if accuracy drops >2%.

4. Production metrics pipeline:

   - Track confidence score distributions over time

   - Monitor escalation rates (sudden spikes indicate extraction problems)

   - Measure end-to-end latency (target: <10s per invoice)

   - Log all agent reasoning for audit trails

5. A/B testing framework: When improving models, route 10% of traffic to new version, compare metrics before full rollout.

Infrastructure requirements: Horizontal scaling with queue-based processing (Redis/RabbitMQ), async Gemini API calls, and caching of PO database lookups to handle 10K invoices within business hours (~7 invoices/second peak).